NEYCO SMART COUNTER POS

Complete System Architecture, UI/UX Guide & Commercial Proposal

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A next-generation, offline-first restaurant management & touch-billing terminal.
Built for high-volume cafes, quick-service restaurants (QSR), food trucks, and fine dine outlets.
Featuring real-time cloud sync, Bluetooth/USB OTG thermal receipt printing, and live analytics.

KEY HIGHLIGHTS

[x] 100% Offline Resilience (Zero Downtime during Internet Outages)
[x] Dual Thermal Printing (Bluetooth SPP & Direct USB OTG Cable)
[x] Non-Blocking Database-First Checkout with Optional Printing
[x] Multi-Tender Split Billing (Cash + UPI / GPay + Card)
[x] Real-time Supabase Cloud Backend & Automated Daily Sales BI

DOCUMENT METADATA

Version: 2.1 Enterprise Ready | Platform: Android / Flutter 3.41 | Backend: Supabase PostgreSQL
Prepared For: Restaurant Owners, General Managers & Operating Teams

1. Executive Summary & Business Value

Why modern food businesses are transitioning from legacy POS systems to Neyco.

Modern food and beverage operations operate in high-pressure environments where speed, reliability, and accuracy directly determine customer satisfaction and profitability. Traditional legacy POS systems often lock restaurants into expensive proprietary hardware (Rs. 40,000+ per terminal), complex on-premise servers, and fragile networking architectures that halt operations when internet or local WiFi fails.

The Neyco Smart Counter POS is engineered to completely eliminate these friction points. Built with Google Flutter and backed by an enterprise-grade Supabase PostgreSQL cloud backend, it delivers lightning-fast touch-screen counter billing, seamless multi-channel payment reconciliation, and rock-solid thermal printing over both wireless Bluetooth and wired USB OTG connections.

Core Competitive Advantages

- **Zero Downtime Guarantee:** Full offline billing with embedded Hive caching. In the event of total network disconnection, cashier operations continue without a hitch. Transactions automatically sync when connection returns.
- **Queue-Busting Speed:** Optimized touch UX allows cashiers to complete a 5-item order with customized add-ons and payment tender in under 4 seconds.
- **Dual Printing Flexibility:** Supports both Bluetooth portable thermal printers and wired USB OTG desktop receipt printers with 58mm (2-inch) and 80mm (3-inch) ESC/POS formatting.
- **Optional Eco-Friendly Printing:** Database-first architecture saves every bill to the cloud immediately while presenting an optional receipt dialog, saving rolls of thermal paper daily.
- **Restaurant Trading Shift Logic:** Intelligent accounting engine groups late-night orders placed past midnight (e.g. up to 3:00 AM) into the same business day shift, eliminating daily sales reporting discrepancies.
- **Low Total Cost of Ownership:** Runs on standard Android tablets and smartphones, slashing initial hardware setup capital by over 75%.

Enterprise Reliability Standard

All transactions receive atomic sequence-guaranteed invoice numbers from the cloud database, preventing invoice collisions across multiple billing counters while maintaining absolute tamper-proof audit trails.

2. End-to-End System Architecture

A multi-layered reactive architecture built for speed, fault tolerance, and security.

Architectural Layers

- **Presentation Layer:** Flutter 3.41 framework rendering at 60 FPS. Implements adaptive layouts tailored for tablets, touch displays, and handheld Android devices.
- **State Management:** Provider & ChangeNotifier reactive state bus. Decouples cart mutations, menu filtering, and payment state from background network IO.
- **Local Persistence Engine:** Hive NoSQL key-value store. Caches complete catalog hierarchies, modifier sets, customer history, and pending offline sync queues with sub-millisecond read latency.
- **Cloud Data Infrastructure:** Supabase Managed PostgreSQL. Features real-time WebSocket replication, Row-Level Security (RLS) policies, and atomic sequence generators.
- **Hardware I/O Driver Layer:** Native ESC/POS byte generator (esc_pos_utils_plus) communicating directly via Android USB Host API and Bluetooth SPP RFCOMM streams.

System Data Flow Matrix

Lifecycle Event	Local Device Action	Cloud Backend Action	Hardware Action
Item Added to Cart	Instant state update in RAM & UI	None (Zero network overhead)	None
Checkout Initiated	Calculates subtotal, discounts & ta	Validates store status	Pre-allocates receipt buffer
Save & Complete	Queues order in Hive cache	Generates invoice # in Postgres	Triggers print preview modal
Print Bill Clicked	Logs print status	Updates 'is_printed' cloud flag	Streams ESC/POS bytes over USB/BT
Offline Operation	Stores order in offline_orders box	Synced on reconnection	Prints receipt with offline badge
Shift Closing (Z-Rep)	Aggregates shift KPIs locally	Audits against Supabase DB	Prints consolidated thermal report

Offline-to-Online Cloud Sync Engine

When internet connectivity drops, orders are cryptographically timestamped and stored in the local SQLite/Hive queue. As soon as the network interface detects active internet, a non-blocking background worker processes and verifies the queue into Supabase without disrupting active cashier billing.

3. Exhaustive UI/UX Screen-by-Screen Guide (Part 1)

High-speed Counter POS and Live Cart Management.

Screen 1: Main POS Counter Terminal (PosScreen)

The primary billing interface is engineered for rapid visual recognition and minimal touch latency. It consists of a top navigation header, category pill carousel, responsive item card grid, and an integrated real-time search engine.

- **Top Navigation Header:** Features real-time store branding, live clock, internet/cloud sync status pill, dynamic printer connection status badge (Green 'Printer Connected' / Red 'No Printer'), and direct access icons to Daily Sales & Settings.
- **Category Selector Bar:** Horizontally scrolling tabs (e.g. All, Burgers, Beverages, Desserts, Combos). Tapping a category instantly filters the menu grid with smooth transition animations.
- **Visual Menu Item Cards:** High-contrast cards displaying item photography, clear typography, bold price badges (in Indian Rupees), and intuitive 'Out of Stock' overlays when an item is unavailable.
- **Instant Search Filter:** Real-time fuzzy search bar allowing cashiers to filter 500+ menu items by typing 2 characters (e.g., 'bu' filters instantly to Burger selections).

Screen 2: Live Cart & Order Control Panel (CartPanel)

Positioned on the right side (tablet layout) or expandable drawer (phone layout), the Cart Panel manages the active order payload before checkout.

- **Order Type Switcher:** Three distinct segment pills: [Dine-In] (prompts table number input), [Takeaway / Parcel] (prompts customer token/name), and [Delivery].
- **Line Item Controllers:** Each cart item displays quantity increment/decrement (+ / -) controls, unit rate, calculated line total, and a 1-tap trash removal action.
- **Item Notes & Modifiers:** Cashiers can attach special cooking instructions (e.g., 'Less Spicy', 'Extra Cheese') that persist to the database and print on the kitchen receipt.
- **Dynamic Discount Engine:** Dedicated discount button supporting either Flat Rs. amount (e.g. Rs. 50 off) or Percentage % (e.g. 10% staff discount), instantly updating the Grand Total.
- **Checkout Actions:** Two prominent high-contrast buttons: 'Quick Cash' for instant 1-tap cash billing, and 'Save & Complete' to open the full payment and split modal.

4. Exhaustive UI/UX Screen-by-Screen Guide (Part 2)

Tender Settlement, Split Payments, and Interactive Receipt Modal.

Screen 3: Payment & Split Settlement (PaymentModal & SplitPaymentModal)

Neyco provides comprehensive payment tender options to handle the full spectrum of customer payment preferences with complete auditability.

- **Single Tender Modal:** Allows one-tap settlement via Cash (with automated change-due calculator showing exact balance to return to the customer), UPI / QR Code (Google Pay, PhonePe, Paytm), or Credit/Debit Card.
- **Split Payment Matrix:** Enables split settlements where customers pay across multiple methods (e.g., a Rs. 1,000 bill paid as Rs. 400 Cash + Rs. 600 UPI).
- **Live Balance Validator:** Displays the live 'Remaining Amount' as inputs are entered. The 'Complete Split' button is safety-locked until the exact bill total is matched, preventing under or over-collection.
- **Transaction Audit Trail:** All split components are individually serialized and stored in the database for daily drawer auditing.

Screen 4: Order Receipt & Non-Blocking Print Modal (OrderReceiptDialog)

Unlike older systems that stall the POS terminal while waiting for physical printer gears to turn, Neyco decouples cloud database storage from physical printing.

- **Database-First Security:** The moment checkout is completed, the order is safely committed to Supabase and assigned an immutable invoice number.
- **Digital Receipt Preview:** A small, elegant modal pops up displaying the restaurant header, invoice number, timestamp, itemized breakdown, and payment badge.
- **Dedicated 'Print Bill' Button:** Tapping 'Print Bill' immediately streams formatted ESC/POS commands to the active Bluetooth or USB thermal printer with a live loading spinner.
- **Dedicated 'Close' Button:** Tapping 'Close' dismisses the modal in 0.1 seconds without printing, preserving paper while ensuring the transaction is 100% recorded.
- **Reprint Capability:** Any past bill can be reprinted at any time from the Daily Sales screen.

5. Exhaustive UI/UX Screen-by-Screen Guide (Part 3)

Daily Sales Business Intelligence & Hardware Configuration.

Screen 5: Daily Sales & Analytics Dashboard (DailySalesScreen)

A business intelligence suite designed for restaurant owners and managers to monitor live operational performance without needing external third-party software.

- **Real-Time KPI Cards:** Displays Gross Sales (Rs.), Net Revenue, Total Completed Orders, Average Order Value (AOV), and Total Discounts Given.
- **Payment Method Breakdown:** Visual revenue split across Cash, UPI / Digital, Card, and Split settlements.
- **Restaurant Trading Shift Logic:** Allows setting custom day cutoff hours (e.g. 3:00 AM) so post-midnight orders are accurately credited to the previous evening's shift.
- **Top-Selling Menu Items:** Ranked list of highest-grossing dishes and beverage volume to optimize kitchen prep.
- **Historical Order Ledger:** Searchable, filterable list of all daily orders with details, invoice numbers, and reprint actions.
- **Thermal Shift Reports (X/Z Reports):** One-tap button to print mid-shift (X-Report) and end-of-day register closing (Z-Report) thermal summary slips.
- **CSV / Excel Data Export:** Download full transaction audit logs formatted for standard accounting software.

Screen 6: Printer Settings & Hardware Hub (PrinterSettingsScreen)

A centralized hardware manager enabling plug-and-play setup for Bluetooth and USB OTG thermal printers.

- **Dual Mode Switcher:** A segmented tab bar switching between [Bluetooth Mode] and [USB Cable (OTG)].
- **Automated Device Discovery:** Scans for bonded Bluetooth printers and connected USB OTG receipt printers in under 2 seconds.
- **Paper Width Customization:** Toggle between 58mm (32 characters/line) and 80mm (42 characters/line) layout engines.
- **Hardware Diagnostics:** Integrated 'Print Test Receipt' button that checks connectivity, font sizes, alignment, and paper cut mechanisms.
- **Store Branding Editor:** Configure Restaurant Name, Address, Phone Number, FSSAI/GST numbers, and customized receipt footer text.

6. Thermal Receipt Layout & ESC/POS Specifications

Pixel-perfect, standards-compliant thermal receipts across 58mm and 80mm formats.

Receipt Formatting Anatomy

The printing engine uses binary ESC/POS commands directly streamed to the thermal head. Text sizes, bold weights, alignments, and dividers are dynamically computed based on active paper width (58mm vs 80mm).

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                    NEYCO RESTAURANT & CAFE
          123 Commercial St, Bangalore
                Ph: +91 98765 43210
 ========================================
 INV: NY-2026-09028    DATE: 20-Sep-2026
 TYPE: DINE-IN (T-04)  TIME: 08:45 PM
 PAYMENT: [ CASH ]
 ----------------------------------------
 ITEM                    QTY   RATE  TOTAL
 ----------------------------------------
 Classic Chicken Burger    2   180.0  360.0
 Peri Peri Fries (Large)   1   120.0  120.0
 Cold Coffee Frappe        2   110.0  220.0
 ----------------------------------------
SUBTOTAL:                        Rs.700.00
 DISCOUNT (10%):                -Rs.70.00
 ========================================
GRAND TOTAL:                     Rs.630.00
 ========================================
           THANK YOU! VISIT AGAIN
         Powered by Neyco Smart POS
```

- **58mm Standard (2-Inch):** 32 columns per line, compact font sizing, optimized spacing for mobile belt printers.
- **80mm Standard (3-Inch):** 42-48 columns per line, expanded item names, wide subtotal alignment, automatic paper-cutting support.

7. Hardware Requirements, TCO & Deployment Roadmap

Commercial proposal and onboarding timeline for new restaurant locations.

Hardware Comparison & Total Cost of Ownership (TCO)

Component	Legacy Proprietary POS	Neyco Smart Counter POS
POS Hardware Terminal	Rs. 35,000 - Rs. 55,000 (Locked)	Rs. 8,000 - Rs. 12,000 (Any Android Tab/Phone)
Thermal Receipt Printer	Rs. 12,000 (Proprietary LAN/Serial)	Rs. 2,500 - Rs. 4,000 (Standard BT/USB)
Local On-Premise Server	Rs. 20,000 + UPS Maintenance	Rs. 0 (Built-in SQLite / Hive Engine)
Annual Maintenance (AMC)	Rs. 15,000 - Rs. 25,000 / year	Zero mandatory maintenance fees
Internet Failure Risk	Operations stop / High loss	100% Offline Continuous Billing
TOTAL INITIAL INVESTMENT	**Rs. 67,000 - Rs. 90,000+**	**Rs. 10,500 - Rs. 16,000 (Over 75% Savings)**

4-Day Rapid Onboarding & Deployment Roadmap

- **Day 1: Menu Digitization:** Cataloging categories, menu items, prices, tax slabs, and modifiers into the Supabase Cloud database.
- **Day 2: Hardware Setup:** Pairing Bluetooth and USB OTG thermal printers, setting receipt branding headers and paper sizes.
- **Day 3: Staff Training:** 30-minute interactive cashier orientation covering billing, split tenders, discounts, and shift closing.
- **Day 4: Live Go-Live:** Operational launch with live cloud monitoring and instant sales synchronization.

Proposal Acceptance & Next Steps

To initiate setup or schedule a live on-site demo with thermal printer hardware testing, please contact the Neyco implementation team. Deployment is completed within 24 hours of menu confirmation.